WORKED SOLUTION

STEP I 2018, Question 5

Pure • Polynomials and Factor Theorem

STEP I 2018, Question 5

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/18-s1-q5

The key idea

"Takes the value 1 at these points" is really a statement about roots: if $P(k) = 1$ then k is a root of $P(x) - 1$, so $x - k$ is a factor of it. Every part of the question becomes easy once we subtract the common value and factorise. In part (iii) the factorisation turns questions about S into questions about writing a small number as a product of four *distinct* integers, and distinctness is what does all the work.

Solution

Part (i)

By the remainder theorem, the remainder on dividing $P(x)$ by $x - k$ is $P(k)$. So we need $P(1) = P(2) = P(3) = P(4) = 1$, that is, $1, 2, 3, 4$ are roots of $P(x) - 1$. Since $P(x) - 1$ has degree at most 4 and these four distinct roots, it is a constant multiple of $(x-1)(x-2)(x-3)(x-4)$. Hence the most general such polynomial is

$$P(x) = 1 + k(x-1)(x-2)(x-3)(x-4), \qquad k \neq 0,$$

where $k \neq 0$ is needed for the degree to be exactly 4.

Part (ii)

Exactly as in part (i), $1, 2, \ldots, N$ are N distinct roots of $P(x) - 1$, which has degree N (because $N \geq 1$, subtracting 1 does not change the degree). So

$$P(x) - 1 = k(x-1)(x-2)\cdots(x-N)$$

for some constant k, and $k \neq 0$ since P has degree $N \geq 1$. Then

$$P(N+1) - 1 = k \cdot N(N-1)\cdots 1 = k\,N! \neq 0,$$

as both k and $N!$ are non-zero. So $P(N+1) \neq 1$.

If $P(N+1) = 2$, then $k\,N! = 1$, so $k = \frac{1}{N!}$. For a positive integer r,

$$\begin{aligned} P(N+r) &= 1 + \frac{(N+r-1)(N+r-2)\cdots r}{N!} \\ &= 1 + \frac{(N+r-1)!}{(r-1)!\,N!} = 1 + \binom{N+r-1}{N}. \end{aligned}$$

(The product has N factors, running down from $N+r-1$ to r.)

We now want $1 + \binom{N+r-1}{N} = N + r$ for every N. Try small r. With $r = 1$ we get $P(N+1) = 2$, which equals $N+1$ only when $N = 1$. With $r = 2$,

$$P(N+2) = 1 + \binom{N+1}{N} = 1 + (N+1) = N+2,$$

so $r = 2$ works whatever N is.

Part (iii)

The four distinct integers a, b, c, d are roots of $S(x) - 2001$, which has degree 4 and leading coefficient 1. So

$$S(x) = 2001 + (x-a)(x-b)(x-c)(x-d).$$

(a) Suppose $S(e) = 2018$ for some integer e. Then

$$(e-a)(e-b)(e-c)(e-d) = 17.$$

The four factors are integers, and they are distinct because a, b, c, d are. None of them is 0, since the product is not.

? **Ask yourself** How can four distinct integers fail to multiply to a prime?

Among distinct non-zero integers, at most two can have absolute value 1 (namely 1 and -1). So at least two of the four factors have absolute value at least 2, and the product of those two has absolute value at least 4. But that product divides 17, so it must be ± 17; since 17 is prime, it cannot be written as a product of two integers each of absolute value at least 2. This contradiction shows that no integer e has $S(e) = 2018$.

(b) Now $S(0) = 2001 + (-a)(-b)(-c)(-d) = 2001 + abcd$, so $S(0) = 2017$ exactly when

$$abcd = 16.$$

Conversely, any four distinct integers with $abcd = 16$ give a valid S (the formula above is monic with integer coefficients), so we must count sets $\{a, b, c, d\}$ of four distinct integers with product 16; each set can be put in increasing order in exactly one way.

Each of a, b, c, d divides 16, so its absolute value is one of $1, 2, 4, 8, 16$, and the four absolute values multiply to 16. Because the integers are distinct, each absolute value

can occur at most twice (as $\pm m$). Writing the absolute values as powers 2^p with the four exponents adding to 4 and no exponent used more than twice, the possibilities are

$$\{1,1,2,8\}, \qquad \{1,1,4,4\}, \qquad \{1,2,2,4\}.$$

(Using 16 would force the other three to be 1, 1, 1, which is not allowed; and $\{2,2,2,2\}$ uses 2 four times.) Now choose signs so that the product is $+16$:

- $\{1,1,2,8\}$: the pair is $1,-1$, contributing -1, so exactly one of 2, 8 is negative. This gives $\{-2,-1,1,8\}$ and $\{-8,-1,1,2\}$.
- $\{1,1,4,4\}$: both pairs are $\pm$, and $(1)(-1)(4)(-4)=16$. This gives $\{-4,-1,1,4\}$.
- $\{1,2,2,4\}$: the pair $2,-2$ contributes -4, so exactly one of 1, 4 is negative. This gives $\{-4,-2,1,2\}$ and $\{-2,-1,2,4\}$.

So there are **5** ways to choose $a<b<c<d$.

Checks

- In part (ii) with $N=1$, $P(x)=x$, so $P(N+r)=1+r$ agrees with $1+\binom{r}{1}$, and $P(3)=3$. With $N=2$, $P(x)=1+\frac{1}{2}(x-1)(x-2)$ gives $P(4)=4$, as claimed for $r=2$.
- For (iii)(b), multiply out each of the five sets: for example $(-2)(-1)(2)(4)=16$. A computer search over all quadruples of distinct integers in $[-16,16]$ finds exactly these five.
- For (iii)(a), note that $2018-2001=17$ is prime; with 2001 replaced by a value giving a difference like $4=(-2)(-1)(1)(2)$, such an e could exist, so primality really matters.

Common slips

- Forgetting $k\neq 0$ in part (i), or writing $P(x)=(x-1)(x-2)(x-3)(x-4)+1$ only, which is not the most general polynomial.
- In part (iii)(b), counting only positive divisors, or forgetting that 1 and -1 can both be used.
- Writing $S(0)=2001-abcd$: there are four minus signs, so the product is $+abcd$.

Beyond STEP

The idea that a polynomial of degree N is determined by its values at $N+1$ points underlies Lagrange interpolation and finite differences; part (ii) is the polynomial $1+\binom{x-1}{N}$ in disguise. Part (iii) is a small instance of a standard tool in number theory: if P has integer coefficients then $m-n$ divides $P(m)-P(n)$, which severely limits the

values a polynomial can take at integers. Davenport's *The Higher Arithmetic* is a good place to read more about divisibility arguments of this kind.

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